Birla Institute of Technology and Science, Pilani
Second Semester 2019-2020
EEE/INSTR F242: Control Systems
Comprehensive Examination

Date: 28.12.2020 **Time: 2 Hrs** **MM: 135**

Q1. In a unity negative feedback control system, the transfer function of plant is $\frac{1}{s(s-1)}$.

(a) Show that the system can not be made stable with a proportional controller (Gain K).

(b) Now a derivative error controller (gain K_d) is also introduced in the system. Is the system stable now? If yes, then what is the range of K and K_d for system to be stable? What would be the value of K_d for system to be marginal stable and corresponding frequency of sustained oscillations?

(c) For the original system, design a controller which is a combination of proportional error controller (gain K) and output derivative controller (gain K_1) to satisfy the following time response specification:
Peak overshoot: 9.4% and Rise time: 0.221 sec. Determine the values of controller gains. [2+6+17]

Q2. The block diagram of a control system is shown in Fig. Q.2. For this system:

(i) Clearly show the region in K-α Plane that can be used without instability arising.

(ii) Draw the neat sketches of root contours for K=11.25 and 18.25. Clearly show the steps involved.

(iii) While drawing the root locus, it is observed that the for certain value of K and α, system has damping ratio of 0.707 and settling time of 1.5 sec for 5% tolerance band corresponding to dominant pole pair. Calculate this value of K and α. [12+15+8]

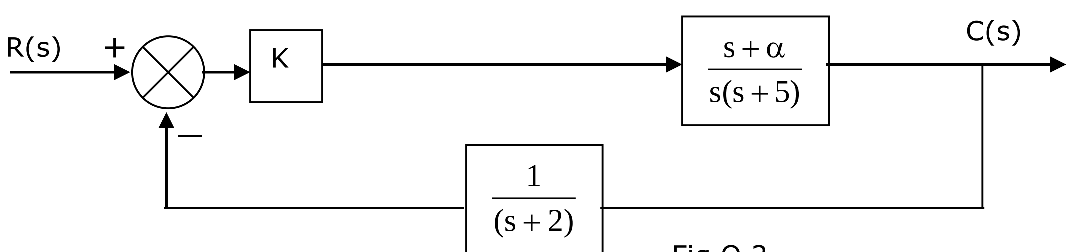
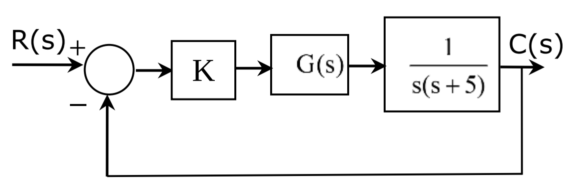


Fig Q.2

Q3. In the block diagram given, G(s) represents the transfer function for which the polar plot is shown. Determine

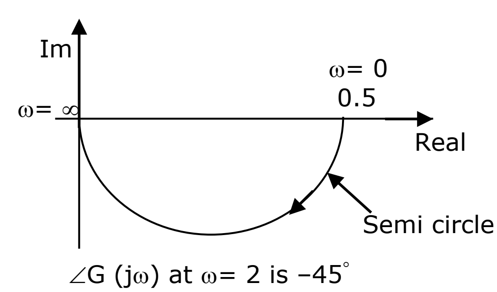


(i) G(s)

(ii) Value of K if phase margin is 35° and corresponding gain margin (calculate mathematically).

(iii) What is the steady state error for a unit ramp input? How can this error be made zero for same input?

(iv) Draw the labeled Bode's asymptotic plot for K=1 in your answer sheet. [10+15+5+10]



Q4. Consider the system shown in Fig. Q4, which is designed to control the liquid level *H* in a cylindrical tank.

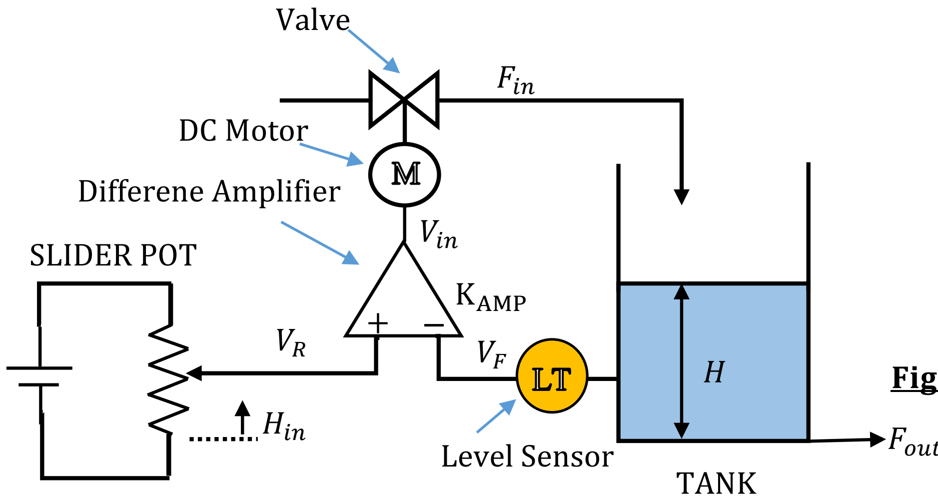


Fig. Q4

A linear Slider POT is used which serves the purpose of feeding reference signal to level control system and has a sensitivity of 0.2 V/cm. The Level sensor (LT) has sensitivity of 0.2 V/cm. An ideal linear difference amplifier is used with a voltage gain (K_{AMP}) of 1 V/V. Further, a motorized valve is used to vary the input flow rate (F_{in}). This arrangement has two parts viz. a DC motor and a valve. The valve is attached to the DC motor shaft and provides 10 liters per min (LPM) for 10° of motor shaft rotation. The DC motor dynamics are given as,

$$G_1(s) = \frac{\omega_R(s)}{V_{in}(s)} = \frac{3.10}{(1+0.1s)}$$

where, ω_R is the angular speed of motor in rad/s and V_{in} is the motor driving voltage in V. The tank contains the liquid with F_{in} (in LPM) and F_{out} (in LPM) as the input and output flow rate respectively. The liquid level *H* (cm) is related with input flow rate using a first order dynamics and has a transfer function as,

$$G_2(s) = \frac{H(s)}{F_{in}(s)} = \frac{2}{(1+0.25s)}$$

For this system,

(i) Determine transfer function $H(s)/H_{in}(s)$

(ii) Sensitivity of the system with respect to change in K_{AMP} for ω=1 rad/s

(iii) Draw the Nyquist plot, determine the gain margin and comment on system stability.

[5+5+25]